Stanford

Weekly update 1 of 5 · week of Nov 9

Data, ground truth and experiment setup

THIS UPDATE IN THREE NUMBERS

292

buildings with an enclosure A1–A3 reference value (Benke et al. 2025)

78

median $kgCO_2e/m^2$ floor area; range 0.5–514

15.5%

of all enclosure carbon comes from XPS insulation alone

Stanford
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Ground truth: enclosure A1–A3 per m^2 of floor area

55,807

material rows: Shell – Enclosure, stage A1–A3

243 + 49

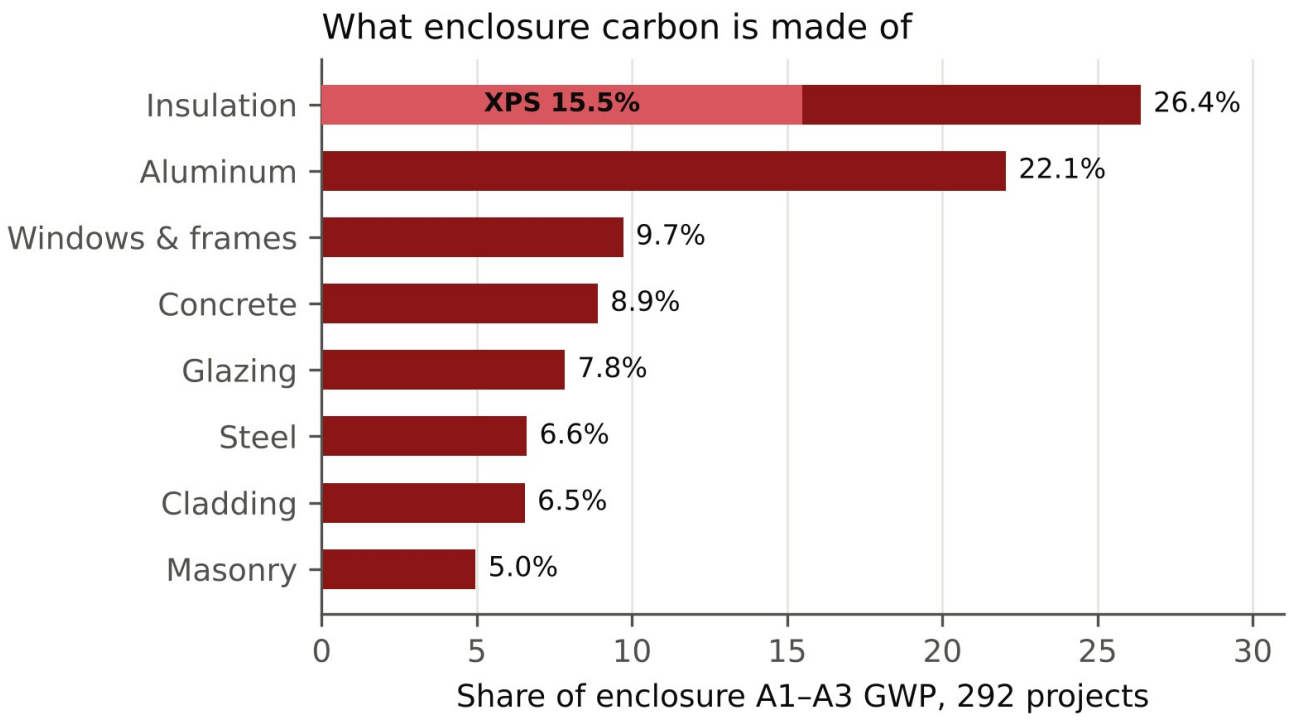
new construction + renovations

19%

median share of whole-building A–C carbon

- **Source:** harmonized North American WBLCA dataset (Benke et al. 2025, CC BY 4.0), 48 early-stage features per project.
- **Normalization:** conditioned floor area; renovated + added GFA for renovations. This reproduces the dataset's own normalization for all 292 projects.
- **Scope:** the enclosure includes windows, doors and roofs. The unit is per m² of floor area, because wall area is missing for most projects (the proposal planned per m² of wall).

What enclosure carbon is made of



- **Insulation (26.4%) and aluminum (22.0%)** dominate.
- **XPS: 43.6 kgCO_2e/kg**, against 1.2 for mineral wool and 3.1 for polyiso: the HFC-blown product that Tally and One Click LCA modelled in the assessment years.
- **149 of 292** projects contain XPS.
- **Data quality:** 10 new-construction projects have no fenestration GWP at all, several with a WWR of 0.5–0.7. They are flagged, and results are reported with and without them.

Climate vs use: use matters more (proposal RQ1)

53.6

median $kgCO_2e/m^2$, multifamily (44 projects)

87.2

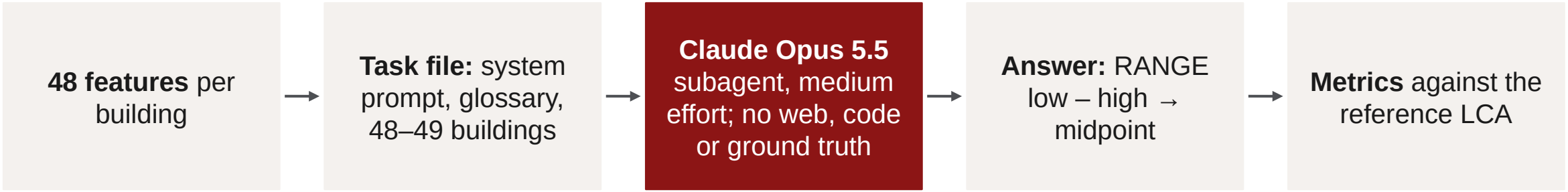
median, office (41)

157.2

median, transportation hubs (9)

- **Use differs significantly** across 9 uses (Kruskal–Wallis $p = 2.7 \times 10^{-7}$). New construction with complete references, 233 projects.
- **Climate does not reach significance** ($p = 0.26$): warm zones 80.6, zone 4 85.9, cold zones 95.1.
- Per m^2 of floor area, use (glazing share, envelope-to-floor ratio) shapes enclosure carbon more than climate (insulation).

LLM experiment protocol



- **3 runs × 292 buildings** = 876 estimates per prompt. Seeded shuffles; identical chunks across prompt versions, so comparisons are paired building by building.
- **Metrics:** ±20% coverage (Wilson CI), RMSE, median APE, median |log(est/ref)|, bias (geometric mean of est/ref), CV across runs.
- **Parser change:** a lower bound of 0 is accepted. Renovations that leave the envelope untouched have reference values near zero, and the course parser would have dropped exactly those answers.

Next update

Coming next

- **Baseline results** with a plain prompt (the model's own knowledge, prompt v1).
- **Error analysis:** where the model under- and overestimates, and why.

Input we'd like

- Is **per m² of floor area** acceptable as the functional unit, instead of per m² of wall?
- Keep the **10 flagged incomplete references** in the main metrics (we report both ways)?